No of Pages : 3 **Course Code : 23MX23**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE 641 004

SEMESTER EXAMINATIONS, JUNE 2024

MCA Semester: 2

23MX23 OBJECT ORIENTED PROGRAMMING USING JAVA

Time : 3 Hours **Maximum Marks:100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Course Outcome : Table

Qn.1	CO1	Qn.2	CO2	Qn.3	CO3	Qn.4	CO4

COs	At the end of learning this course, students will be able to
CO1	Comprehend object oriented concepts and fundamentals of Java for problem solving
CO2	Write applications using Object Oriented features with proper exception handling mechanisms
CO3	Write programs using streams and multithreading
CO4	Design and develop applications with user interfacing, Collections framework and database connectivity

		Question		**(BTL)**
1.	a)	State what is an object with an example.	(3)	(L1)
	b) i)	Exemplify the use of Arraylist. Consider a program written using an array A of size 50. Now suppose you want to change the program so that A is an ArrayList. What changes must or should be made to the program? Your goal is to have a correctly functioning program that follows good programming practice, while minimizing changes to the code.	(5)	L3
	ii)	Implement the static method mode, which returns the value that occurs most often in an array of integers. You may assume the array has at least one element, and all of the values in the array are in the range 0- 100. If there is more than one value that occurs most often, return the smallest of such value.	(5)	(L3)
	c)	i) State the advantages of Object-Oriented Programming features in Java.	(4)	(L4)
		ii) Using abstraction concept, design a program for employee class contains the variable (ID, name, mobile-no and salary) and do the following operations: • Initial values for 4 Employee using ArrayList concept. • Display the information of Employee who has name = “Kavin”.	(8)	

2.	a)	Consider the following code:
		public final class Test {
		final int x=10;
		... // code continues from here
		In this context, what does the word final signify about the class Test.
			(3)	(L1)

b) i) Distinguish **String** and **StringBuffer** class. Determine the output for the following Java code and provide the explanation. What will happen if the following code is executed? Justify it. (5) (L3)

```
String str1= new String("Data");
String str2= "Data";
If(str1== str2)
   System.out.println("The strings are equal");
else
   System.out.println(" The strings are not equal");
```

ii) Illustrate the significant use of exception handling and analyse the following program. (5) (L3)

```
class A {
static void my() throws ArithmeticException {
                System.out.print("A");
                throw new ArithmeticException("A");
   }
public static void main (String args []) {
    try {
         my();
    }
    catch (Exception e) {
         System.out.print("B");
    }
    finally {
         System.out.print("C");
    }
   }
   }
```

c) i) Create a general class ThreeDObject and derive the classes Box, Cube, Cylinder and Cone from it. The class ThreeDObject has methods wholeSurfaceArea ( ) and volume ( ). Override these two methods in each of the derived classes to calculate the volume and whole surface area of each type of three-dimensional objects. The dimensions of the objects are to be taken from the users and passed through the respective constructors of each derived class. Write a main method to test these classes. (12) (L5)

(OR)

ii) 1) Explain how does the inheritance help in achieving run time polymorphism in java. Explain the need of Abstract class and Abstract method with example. (5) (L2)

2) Design an abstract class "Shape" with members width, height and abstract function voidarea(). Create classes Square, Triangle and Rectangle that inherit Shape and override function area with custom definition. Write a code to create objects for classes and compute its area. (7) (L5)

3. a) Compare extending thread class and implementing runnable interface. Is it better to extend thread or implement Runnable? (3) (L2)

b) i) Write a program using the File Reader and File Writer to copy the contents of the file “test.in” to the file “answer.out”. (5) (L3)

ii) What will be the output of the program? Analyse it. (5) (L3)

```
class MyThread extends Thread {
public static void main(String [ ] args) {
MyThread t = new MyThread();
t.start();
System.out.print("one. ");
t.start();
System.out.print("two. ");
}
public void run() {
System.out.print("Thread ");
}}
```

c) i) Distinguish the Reader/Writer class hierarchy and the InputStream/OutputStream class hierarchy? Can we say the file streams in java works on the principles of object serialization? Justify your answer. (12) (L3)

(OR)

ii) Illustrate threads life cycle with relevant methods call with suitable diagram.

4. a) State the need of garbage collection. Which part of the memory is involved in Garbage Collection? (3) (L1)

b) i) With relevant examples, discuss about Set, Vector and Hash map in Java Utility Class. (5) (L2)

ii) Paraphrase various Event classes and listener interfaces for handling events generated in an Applet. (5) (L2)

c) Exemplify the classes and interfaces available in the collection framework. Suppose there are ten cars parked in a parking area. A token number is given to the owner of each car, and the same number is stamped on the car by the parking caretaker. The parked cars have the following sequence of numbers 10, 20, 30, 40, 50, 60, 70, 80, 90, 100 and are parked in the same order as previously mentioned. (L5)

The caretaker reverses the order of a portion of the sequence of cars and arranges the vehicles in order 80, 70, 60, 50, 40, 30, 20, 10, 90, 100.

Write an efficient JAVA program using collection class to find the order of cars after placing a portion of cars in the reverse order. (12)

CO : COURSE OUTCOME; BTL : BLOOM’S TAXONOMY LEVEL

/END/

FD/RL